

MSDM 5058 Information Science
Assignment 3 (due 28th March, 2026)

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(1) Conditioning reduces entropy

In the lecture, it is stated that “conditioning reduces entropy”, i.e., $H(X|Y) \leq H(X)$ with equality if and only if X is independent of Y . However, it can happen that $H(X|Y = y) < H(X)$ or $H(X|Y = y) > H(X)$. We here study an example to see if this is the case. The following table shows the joint distribution $p(x, y) = P(X = x, Y = y)$ of X and Y .

$p(x, y)$	$x = 1$	$x = 2$	$x = 3$	$x = 4$
$y = 1$	1/8	1/16	1/32	1/32
$y = 2$	1/16	1/8	1/32	1/32
$y = 3$	1/16	1/16	1/16	1/16
$y = 4$	1/4	0	0	0

- (a) What are the marginal probabilities $P(X)$ and $P(Y)$?
- (b) What is the joint entropy $H(X, Y)$?
- (c) What are the marginal entropies $H(X)$ and $H(Y)$?
- (d) For each value of y , what is the conditional entropy $H(X|Y = y)$?
- (e) For what values of y can one have
 - (i) ... $H(X|Y = y) < H(X)$?
 - (ii) ... $H(X|Y = y) > H(X)$?
- (f) What are the conditional entropies $H(X|Y)$ and $H(Y|X)$?
- (g) Compute the mutual information $I(X; Y)$ and verify that

$$I(X; Y) = H(X) - H(X|Y) = H(Y) - H(Y|X)$$

Solution:

- (a) $P(X = 1) = 1/2$, $P(X = 2) = 1/4$, $P(X = 3) = 1/8$, $P(X = 4) = 1/8$,
 $P(Y = 1) = 1/4$, $P(Y = 2) = 1/4$, $P(Y = 3) = 1/4$, $P(Y = 4) = 1/4$
- (b) $H(X, Y) = -\sum_{x,y} p(x, y) \log_2 p(x, y)$

$$= \frac{1}{4} \log_2 4 + \frac{1}{4} \log_2 8 + \frac{3}{8} \log_2 16 + \frac{1}{8} \log_2 32 = \frac{27}{8} \text{ bits}$$

$$(c) \begin{cases} H(X) = \frac{1}{2} \log_2 2 + \frac{1}{4} \log_2 4 + \frac{2}{8} \log_2 8 = \frac{7}{4} \text{ bits} \\ H(Y) = 4 \times \frac{1}{4} \log_2 4 = 2 \text{ bits} \end{cases}$$

(d) The conditional probabilities $p(x|y) = P(X = x|Y = y)$ are

$p(x y)$	$x = 1$	$x = 2$	$x = 3$	$x = 4$
$y = 1$	$1/2$	$1/4$	$1/8$	$1/8$
$y = 2$	$1/4$	$1/2$	$1/8$	$1/8$
$y = 3$	$1/4$	$1/4$	$1/4$	$1/4$
$y = 4$	1	0	0	0

$$\text{Therefore, } \begin{cases} H(X|Y = 1) = \frac{1}{2} \log_2 2 + \frac{1}{4} \log_2 4 + \frac{2}{8} \log_2 8 = \frac{7}{4} \text{ bits} \\ H(X|Y = 2) = \frac{1}{4} \log_2 4 + \frac{1}{2} \log_2 2 + \frac{2}{8} \log_2 8 = \frac{7}{4} \text{ bits} \\ H(X|Y = 3) = \frac{4}{4} \log_2 4 = 2 \text{ bits} \\ H(X|Y = 4) = 1 \log_2 1 = 0 \end{cases}$$

(e) Compare the results in (c) and (d), we get (i) $H(X|Y = 4) < H(X)$ and (ii) $H(X|Y = 3) > H(X)$.

(f) $H(X|Y) = -\sum_y P(Y = y) H(X|Y = y)$

$$= \frac{1}{4} \left[H(X|Y = 1) + H(X|Y = 2) + H(X|Y = 3) + H(X|Y = 4) \right] = \frac{11}{8} \text{ bits}$$

Similarly, one can get $H(Y|X) = \frac{13}{8} \text{ bits}$

$$(g) \begin{cases} I(X; Y) = H(X) + H(Y) - H(X, Y) = \frac{7}{4} + 2 - \frac{27}{8} = \frac{3}{8} \text{ bits} \\ H(X) - H(X|Y) = \frac{7}{4} - \frac{11}{8} = \frac{3}{8} \text{ bits} \\ H(Y) - H(Y|X) = 2 - \frac{13}{8} = \frac{3}{8} \text{ bits} \end{cases}$$

(2) Principle of maximum entropy

(a) Given a six-face dice, use the principle of maximum entropy to calculate the best guess of the distribution of its six faces if ...

- (i) ... no other information is given.
- (ii) ... its average is known to be $25/6$.

(b) A gas system has a fixed temperature T . Let $\{p_1, p_2, \dots, p_n\}$ be the probabilities for the system to have energy $\{\varepsilon_1, \varepsilon_2, \dots, \varepsilon_n\}$ respectively, where n is the number of energy levels, and let the mean value of energy be $\bar{\varepsilon}$. Show that the Maxwell-Boltzmann distribution

$$p_i = \frac{e^{-\frac{\varepsilon_i}{kT}}}{\sum_{i=1}^n e^{-\frac{\varepsilon_i}{kT}}}$$

maximizes the Shannon entropy

$$-\sum_{i=1}^n p_i \log p_i$$

subject to

$$\sum_{i=1}^n p_i = 1 \quad \text{and} \quad \sum_{i=1}^n p_i \varepsilon_i = \bar{\varepsilon}$$

while k is the Boltzmann constant.

Solution:

- (a) Refer to Lecture 9, p.74 – 78. Let p_i be the probability of the i th face. (i) With no other information, the best guess is a uniform distribution with $p_i = 1/6$. (ii) By knowing the average, we have one more constraint, and the Lagrangian becomes

$$J(p) = -\sum_{i=1}^6 p_i \log p_i + \lambda_0 \left(\sum_{i=1}^6 p_i - 1 \right) + \lambda_1 \left(\sum_{i=1}^6 i p_i - \frac{25}{6} \right)$$

Differentiating it with respect to p_i and setting the derivative to zero, we get the condition of maximum.

$$\frac{\partial}{\partial p_i} J(p) = -1 - \log p_i + \lambda_0 + \lambda_1 i = 0 \quad \Rightarrow \quad p_i = e^{-1+\lambda_0+\lambda_1 i}$$

Therefore,

$$\begin{aligned} 1 &= \sum_{i=1}^6 p_i = e^{-1+\lambda_0} \sum_{i=1}^6 e^{\lambda_1 i} \quad \Rightarrow \quad e^{-1+\lambda_0} = \frac{1}{\sum_{i=1}^6 e^{\lambda_1 i}} \\ \Rightarrow \quad \frac{25}{6} &= \sum_{i=1}^6 i p_i = e^{-1+\lambda_0} \sum_{i=1}^6 i e^{\lambda_1 i} = \frac{\sum_{i=1}^6 i e^{\lambda_1 i}}{\sum_{i=1}^6 e^{\lambda_1 i}} \end{aligned}$$

Finally, we can numerically obtain $\lambda_1 = 0.23634$, $\lambda_0 = -1.6990$, and $p_i = \{0.085204, 0.10792, 0.13669, 0.17313, 0.21929, 0.27776\}$.

- (b) The Lagrangian of the system is given by

$$J(p) = -\sum_{i=1}^n p_i \log p_i + \lambda_0 \left(\sum_{i=1}^n p_i - 1 \right) + \lambda_1 \left(\sum_{i=1}^n p_i \varepsilon_i - \bar{\varepsilon} \right)$$

Differentiating it with respect to p_i and setting the derivative to zero, we get the condition of maximum.

$$\frac{\partial}{\partial p_i} J(p) = -1 - \log p_i + \lambda_0 + \lambda_1 \varepsilon_i = 0 \quad \Rightarrow \quad p_i = \frac{e^{\lambda_1 \varepsilon_i}}{e^{1-\lambda_0}}$$

Using $1 = \sum_{i=1}^n p_i$, we get

$$p_i = \frac{e^{\lambda_1 \varepsilon_i}}{\sum_{i=1}^n e^{\lambda_1 \varepsilon_i}}$$

which becomes the Boltzmann distribution once we substitute $\lambda_1 = -\frac{1}{kT}$. This is possible because the temperature T is also fixed as a constant.

(3) Differential entropy

In the lecture, you learnt how to compute the entropy of discrete random variables. In this question, you are asked to compute the differential entropy of continuous random variables and study its properties. Given a continuous random variable X with a probability density $f(x)$, its differential entropy is defined as

$$H(X) = - \int f(x) \log f(x) dx$$

Let a , λ , and σ be some constants. What is $H(X)$ if ...

- (a) ... X is uniformly distributed between 0 and a but is 0 elsewhere? What happens to $H(X)$ if $a < 1$?
- (b) ... X follows a normal distribution $f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-x^2/2\sigma^2}$?
- (c) ... X follows an exponential distribution $f(x) = \lambda e^{-\lambda x}$?
- (d) ... X follows a Laplace distribution $f(x) = \frac{1}{2} \lambda e^{-\lambda|x|}$?

Solution:

(a)

$$H(X) = - \int_0^a \frac{1}{a} \log \frac{1}{a} dx = \log a$$

When $a < 1$, $H(X)$ becomes negative. This shows that while entropies of discrete random variables are always larger than or equal to zero, differential entropies can take negative values.

(b) As shown in the solution of Assignment 2,

$$H(X) = - \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{x^2}{2\sigma^2}} \left[-\frac{x^2}{2\sigma^2} - \ln \sqrt{2\pi\sigma^2} \right] dx = \frac{1}{2} + \frac{1}{2} \ln(2\pi\sigma^2)$$

(c)

$$\begin{aligned} H(X) &= - \int_0^{\infty} \lambda e^{-\lambda x} (\ln \lambda - \lambda x) dx \\ &= (\ln \lambda) \int_0^{\infty} \underbrace{(-\lambda e^{-\lambda x}) dx}_{d(e^{-\lambda x})} - \lambda \int_0^{\infty} x \underbrace{(-\lambda e^{-\lambda x}) dx}_{d(e^{-\lambda x})} \\ &= (\ln \lambda) \underbrace{[e^{-\lambda x}]_0^{\infty}}_{-1} - \lambda \left(\underbrace{[x e^{-\lambda x}]_0^{\infty}}_0 - \int_0^{\infty} e^{-\lambda x} dx \right) \\ &= -\ln \lambda - \int_0^{\infty} (-\lambda e^{-\lambda x}) dx = -\ln \lambda + 1 \end{aligned}$$

(d)

$$\begin{aligned} H(X) &= - \int_{-\infty}^{\infty} \frac{1}{2} \lambda e^{-\lambda|x|} \left(\ln \frac{\lambda}{2} - \lambda|x| \right) dx \\ &= -\frac{1}{2} \int_{-\infty}^{\infty} \lambda e^{-\lambda|x|} (\ln \lambda - \lambda|x|) dx + \frac{\ln 2}{2} \int_{-\infty}^{\infty} \lambda e^{-\lambda|x|} dx \\ &= \underbrace{- \int_0^{\infty} \lambda e^{-\lambda} (\ln \lambda - \lambda x) dx}_{-\ln \lambda + 1} - (\ln 2) \underbrace{\int_0^{\infty} (-\lambda e^{-\lambda x}) dx}_{-1} \\ &= -\ln \lambda + 1 + \ln 2 = -\ln \frac{\lambda}{2} + 1 \end{aligned}$$

(4) Relative entropy

In the lecture, you learnt that relative entropy is non-negative from probabilistic arguments. You can also prove this divergence inequality in a straightforward way.

(a) Begin by considering the following function

$$f(a) = \log a - a + 1$$

Show that for $a > 0$, $f(a) \leq 0$ and the equality holds if and only if $a = 1$.

(b) Suppose a random variable X can take any value x . By using the result in (a), show that the relative entropy

$$D(p \parallel q) = \sum_x p(x) \log \frac{p(x)}{q(x)}$$

is non-negative.

- (c) We know that $D(p \parallel q)$ is asymmetric with respect to p and q , i.e. $D(p \parallel q) \neq D(q \parallel p)$, but this is not always the case. For example, consider a binary communication channel which uses an alphabet $X = \{0, 1\}$. The probability for its source to send $X = x$ is $p(x)$, while the probability for its target to receive $X = x$ is $q(x)$.

- (i) If $p(0) = 0.3$ and $q(0) = 0.7$, what are $D(p \parallel q)$ and $D(q \parallel p)$?
(ii) When does $D(p \parallel q) = D(q \parallel p)$ happen for this channel?

Solution:

- (a) The first and second derivatives of $f(a)$ are $f'(a) = \frac{1}{a} - 1$ and $f''(a) = -\frac{1}{a^2}$.

The first derivative has a zero when $a = 1$, and the second derivative is always smaller than zero, so $f(x)$ attains its maximum at $a = 1$. Since $f(1) = 0$, $f(a) \leq 0$.

- (b) Notice that $f(a) \leq 0$ implies $-\log a \geq 1 - a$.

$$\begin{aligned} D(p \parallel q) &= \sum_x p(x) \log \frac{p(x)}{q(x)} = \sum_x p(x) \left[-\log \frac{q(x)}{p(x)} \right] \\ &\geq \sum_x p(x) \left[1 - \frac{q(x)}{p(x)} \right] = \sum_x p(x) - \sum_x q(x) = 0 \\ &\therefore D(p \parallel q) \geq 0 \end{aligned}$$

- (c)

$$(i) \quad \begin{cases} D(p \parallel q) = 0.3 \log \left(\frac{0.3}{0.7} \right) + 0.7 \log \left(\frac{0.7}{0.3} \right) \\ D(q \parallel p) = 0.7 \log \left(\frac{0.7}{0.3} \right) + 0.3 \log \left(\frac{0.3}{0.7} \right) \end{cases} \text{ Therefore, } D(p \parallel q) = D(q \parallel p).$$

If base 2 is used, their value is 0.489 bits.

- (ii) For $D(p \parallel q) = D(q \parallel p)$, we should have

$$\begin{aligned} &p(0) \log \frac{p(0)}{q(0)} + [1 - p(0)] \log \frac{1 - p(0)}{1 - q(0)} \\ &= q(0) \log \frac{q(0)}{p(0)} + [1 - q(0)] \log \frac{1 - q(0)}{1 - p(0)} \end{aligned}$$

It holds when $q(x) = p(x)$ or $q(x) = 1 - p(x)$.